



Basic Information

Experiment No: 01

Date: 12/09/2026

Title: Implementation of a Menu-Driven Java Program Using Classes, Methods, Decision-Making and Looping Statements



Objective / Aim

To write a Java program that demonstrates the use of:

- `Scanner` class for user input.
- Classes and objects.
- Methods and method invocation.
- `if-else` and nested `switch` statements.
- `for` and `while` loops.
- Arithmetic and relational operators.
- A menu-driven program structure.
- Basic mathematical operations such as factorial, prime checking, maximum of two numbers, and area of a circle.

The program is divided into four parts:

- **Part A:** Student information and percentage calculation.
 - **Part B:** Even/Odd, largest of three numbers, grade calculation, and day of the week.
 - **Part C:** Multiplication table, numbers from 1 to N, sum of natural numbers, and Fibonacci series.
 - **Part D:** Factorial, prime number checking, maximum of two numbers, and area of a circle.
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System Requirements

Hardware

- Standard PC/Laptop
- Core i3/i5 Processor or equivalent
- Minimum 4 GB RAM
- Keyboard and Monitor

Software

- Operating System: Windows/Linux/macOS
- Java Development Kit: JDK 17 or JDK 21
- Text Editor or IDE:

- VS Code
 - IntelliJ IDEA
 - Eclipse
 - NetBeans
-

Algorithm / Logic Description

Step 1: Start the Program

Import the `Scanner` class and create a `Scanner` object to accept input from the user.

Step 2: Create the `PartD` Class

Create a class named `PartD` containing the following methods:

1. `factorial(int n)` - calculates the factorial of a number using a `for` loop.
2. `prime(int n)` - checks whether a number is prime.
3. `max(int a, int b)` - finds the maximum of two numbers.
4. `areaOfCircle(double r)` - calculates the area of a circle using `Math.PI`.

Step 3: Create the Main Class

Create the public class `Task1` containing the `main()` method.

Step 4: Display the Main Menu

Display the following menu:

1. Part A
2. Part B
3. Part C
4. Part D
5. Exit

Accept the user's choice using `Scanner`.

Step 5: Implement Part A

If the user selects Part A:

1. Read the student's name.
2. Read the student's roll number.
3. Read marks in four subjects.
4. Calculate total marks.
5. Calculate percentage using:

`Percentage = Total Marks / 4.0` 6. Display the student's information.

Step 6: Implement Part B

Display the Part B menu.

Even or Odd

Read a number and use the modulus operator `%` to determine whether it is even or odd.

Largest of Three Numbers

Read three numbers and use `if-else` conditions to determine the largest number.

Grade

Read the percentage and assign grades according to the following conditions:

- 90 or above → Grade A
- 80–89 → Grade B
- 70–79 → Grade C
- 60–69 → Grade D
- Below 60 → Grade E

Days of a Week

Read a number from 1 to 7 and use a `switch` statement to display the corresponding day.

Step 7: Implement Part C

Display the Part C menu.

Multiplication Table

Read a number and use a `for` loop to print its multiplication table from 1 to 10.

Numbers from 1 to N

Read `N` and print numbers from 1 through `N`.

Sum of N Natural Numbers

Read `N`, use a `for` loop, and calculate the sum of the first `N` natural numbers.

Fibonacci Series

Read a number and use a `while` loop to generate Fibonacci numbers up to the given limit.

Step 8: Implement Part D

Create an object of the `PartD` class.

Use the selected option to call the appropriate method:

- `factorial()`
- `prime()`
- `max()`
- `areaOfCircle()`

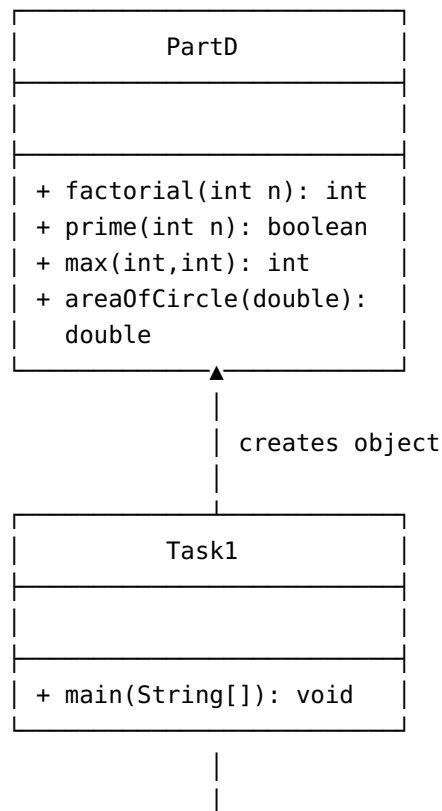
Step 9: Repeat the Main Menu

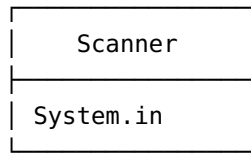
The main menu is executed repeatedly using a `do-while` loop until the user selects `0`.

Step 10: Exit

When the user enters `0`, display `Exiting...`, close the `Scanner`, and terminate the program.

UML Class Diagram





Class Description

PartD

Contains methods for factorial calculation, prime-number checking, finding the maximum of two numbers, and calculating the area of a circle.

Task1

Contains the `main()` method and controls the complete menu-driven program.

Relationship:

`Task1` creates and uses an object of the `PartD` class.



Source Code

Save the program as `Task1.java` because `Task1` is the public class.

```
import java.util.Scanner;

class PartD {

    int factorial(int n) {
        int fact = 1;

        for (int i = 1; i <= n; i++) {
            fact *= i;
        }

        return fact;
    }

    boolean prime(int n) {
        if (n <= 1) {
            return false;
        }
    }
}
```

```

        for (int i = 2; i < n; i++) {
            if (n % i == 0) {
                return false;
            }
        }

        return true;
    }

    int max(int a, int b) {
        return (a > b) ? a : b;
    }

    double areaOfCircle(double r) {
        return Math.PI * r * r;
    }
}

public class Task1 {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
        int choice;

        do {

            System.out.println("\n===== MAIN MENU =====");
            System.out.println("1. Part A");
            System.out.println("2. Part B");
            System.out.println("3. Part C");
            System.out.println("4. Part D");
            System.out.println("0. Exit");

            System.out.print("Enter your choice: ");
            choice = sc.nextInt();

            switch (choice) {

                // ===== PART A =====
                case 1:

                    sc.nextLine();

                    System.out.println("Enter the name of student:");
                    String name = sc.nextLine();

                    System.out.println("Enter the rollno. of student:");

```

```

int rollno = sc.nextInt();

System.out.println(
    "Enter the marks of student in four subjects:"
);

int S1 = sc.nextInt();
int S2 = sc.nextInt();
int S3 = sc.nextInt();
int S4 = sc.nextInt();

int totalmarks = S1 + S2 + S3 + S4;
double percentage = totalmarks / 4.0;

System.out.println("\n===== STUDENT INFORMATION =====");
System.out.println("Name: " + name);
System.out.println("RollNo.: " + rollno);
System.out.println("Total marks: " + totalmarks);
System.out.println("Percentage: " + percentage + "%");

break;

// ===== PART B =====
case 2:

    System.out.println("\n===== PART B =====");
    System.out.println("1. Even or Odd");
    System.out.println("2. Largest of 3 Numbers");
    System.out.println("3. Grade");
    System.out.println("4. Days of a Week");

    System.out.print("Enter your choice: ");
    int partBChoice = sc.nextInt();

    switch (partBChoice) {

        case 1:

            System.out.println("Enter a number:");
            int num = sc.nextInt();

            if (num % 2 == 0) {
                System.out.println(
                    num + " is an even number"
                );
            } else {
                System.out.println(
                    num + " is an odd number"
                );
            }
        }
    }

```

```

        );
    }

    break;

case 2:

    System.out.println("Enter 3 numbers:");

    int a = sc.nextInt();
    int b = sc.nextInt();
    int c = sc.nextInt();

    if (a >= b && a >= c) {
        System.out.println(
            a + " is the largest number"
        );
    } else if (b >= a && b >= c) {
        System.out.println(
            b + " is the largest number"
        );
    } else {
        System.out.println(
            c + " is the largest number"
        );
    }

    break;

case 3:

    System.out.println("Enter the percentage:");
    double per = sc.nextDouble();

    if (per >= 90) {
        System.out.println("Grade A");
    } else if (per >= 80) {
        System.out.println("Grade B");
    } else if (per >= 70) {
        System.out.println("Grade C");
    } else if (per >= 60) {
        System.out.println("Grade D");
    } else {
        System.out.println("Grade E");
    }

    break;

```

```

case 4:

    System.out.println(
        "Enter the Day number (1-7):"
    );

    int day = sc.nextInt();

    switch (day) {

        case 1:
            System.out.println("Monday");
            break;

        case 2:
            System.out.println("Tuesday");
            break;

        case 3:
            System.out.println("Wednesday");
            break;

        case 4:
            System.out.println("Thursday");
            break;

        case 5:
            System.out.println("Friday");
            break;

        case 6:
            System.out.println("Saturday");
            break;

        case 7:
            System.out.println("Sunday");
            break;

        default:
            System.out.println(
                "Enter a valid day number"
            );
    }

    break;

default:
    System.out.println(

```

```

        "Invalid Part B choice."
    );
}

break;

// ===== PART C =====
case 3:

    System.out.println("\n===== PART C =====");
    System.out.println("1. Multiplication Table");
    System.out.println("2. Numbers from 1 to N");
    System.out.println("3. Sum of N Natural Numbers");
    System.out.println("4. Fibonacci Series");

    System.out.print("Enter your choice: ");
    int partCChoice = sc.nextInt();

    switch (partCChoice) {

        case 1:

            System.out.println(
                "Enter a number for multiplication table:"
            );

            int tableNum = sc.nextInt();

            System.out.println(
                "Multiplication table of "
                + tableNum + ":"
            );

            for (int i = 1; i <= 10; i++) {
                System.out.println(
                    tableNum + " x " + i
                    + " = " + (tableNum * i)
                );
            }

            break;

        case 2:

            System.out.println("Enter a number N:");
            int N = sc.nextInt();

            System.out.println(

```

```

        "Numbers from 1 to " + N + ":"
    );

    for (int i = 1; i <= N; i++) {
        System.out.println(i);
    }

    break;

case 3:

    System.out.println("Enter a number N:");
    int N1 = sc.nextInt();

    int sum = 0;

    for (int i = 1; i <= N1; i++) {
        sum += i;
    }

    System.out.println(
        "Sum of first " + N1
        + " natural numbers: " + sum
    );

    break;

case 4:

    System.out.println("Enter a number:");
    int N2 = sc.nextInt();

    int x = 0;
    int y = 1;

    System.out.println(
        "Fibonacci Series up to "
        + N2 + ":"
    );

    while (x <= N2) {

        System.out.print(x + " ");

        int next = x + y;
        x = y;
        y = next;
    }

```

```

        System.out.println();

        break;

    default:
        System.out.println(
            "Invalid Part C choice."
        );
    }

    break;

// ===== PART D =====
case 4:

    System.out.println("\n===== PART D =====");

    PartD obj = new PartD();

    System.out.println("1. Factorial");
    System.out.println("2. Prime");
    System.out.println("3. Maximum of Two Numbers");
    System.out.println("4. Area of Circle");

    System.out.print("Enter your choice: ");
    int partDChoice = sc.nextInt();

    switch (partDChoice) {

        case 1:

            System.out.println("Enter a number:");
            int N3 = sc.nextInt();

            int fact = obj.factorial(N3);

            System.out.println(
                "Factorial of " + N3
                + " is " + fact
            );

            break;

        case 2:

            System.out.println("Enter a number:");
            int N4 = sc.nextInt();

```

```

        boolean isPrime = obj.prime(N4);

        if (isPrime) {
            System.out.println("Prime Number");
        } else {
            System.out.println(
                "Not a Prime Number"
            );
        }

        break;

    case 3:

        System.out.println("Enter 1st number:");
        int num1 = sc.nextInt();

        System.out.println("Enter 2nd number:");
        int num2 = sc.nextInt();

        System.out.println(
            "Maximum = "
            + obj.max(num1, num2)
        );

        break;

    case 4:

        System.out.println("Enter radius:");
        double r = sc.nextDouble();

        System.out.println(
            "Area = "
            + obj.areaOfCircle(r)
        );

        break;

    default:
        System.out.println(
            "Invalid Part D choice."
        );
}

break;

```

```
// ===== EXIT =====  
case 0:  
  
    System.out.println("Exiting...");  
  
    break;  
  
// ===== INVALID =====  
default:  
  
    System.out.println(  
        "Invalid choice. Please try again."  
    );  
}  
  
} while (choice != 0);  
  
sc.close();  
}  
}
```

Compilation & Execution Commands

Step 1: Save the File

Save the source code exactly as:

```
Task1.java
```

The filename must match the public class name `Task1`.

Step 2: Open Terminal / Command Prompt

Navigate to the folder containing `Task1.java`.

Step 3: Compile the Program

```
javac Task1.java
```

If there are no compilation errors, Java creates the corresponding `.class` files.

Step 4: Run the Program

```
java Task1
```

Complete Command Sequence

```
javac Task1.java  
java Task1
```



Output Screen

Sample Output – Main Menu

```
===== MAIN MENU =====  
1. Part A  
2. Part B  
3. Part C  
4. Part D  
0. Exit  
Enter your choice:
```

Sample Output – Part A

```
===== MAIN MENU =====  
1. Part A  
2. Part B  
3. Part C  
4. Part D  
0. Exit  
Enter your choice: 1  
  
Enter the name of student:  
Rahul  
  
Enter the rollno. of student:  
101  
  
Enter the marks of student in four subjects:  
85  
90  
80
```

95

===== STUDENT INFORMATION =====

Name: Rahul

RollNo.: 101

Total marks: 350

Percentage: 87.5%

Sample Output – Part B: Even or Odd

===== MAIN MENU =====

1. Part A

2. Part B

3. Part C

4. Part D

0. Exit

Enter your choice: 2

===== PART B =====

1. Even or Odd

2. Largest of 3 Numbers

3. Grade

4. Days of a Week

Enter your choice: 1

Enter a number:

25

25 is an odd number

Sample Output – Part C: Multiplication Table

===== MAIN MENU =====

1. Part A

2. Part B

3. Part C

4. Part D

0. Exit

Enter your choice: 3

===== PART C =====

1. Multiplication Table

2. Numbers from 1 to N

3. Sum of N Natural Numbers

4. Fibonacci Series

Enter your choice: 1

Enter a number for multiplication table:

5

Multiplication table of 5:

$5 \times 1 = 5$

$5 \times 2 = 10$

$5 \times 3 = 15$

$5 \times 4 = 20$

$5 \times 5 = 25$

$5 \times 6 = 30$

$5 \times 7 = 35$

$5 \times 8 = 40$

$5 \times 9 = 45$

$5 \times 10 = 50$

Sample Output – Part D: Factorial

===== MAIN MENU =====

1. Part A

2. Part B

3. Part C

4. Part D

0. Exit

Enter your choice: 4

===== PART D =====

1. Factorial

2. Prime

3. Maximum of Two Numbers

4. Area of Circle

Enter your choice: 1

Enter a number:

5

Factorial of 5 is 120

Sample Output – Exit

===== MAIN MENU =====

1. Part A

2. Part B

3. Part C

4. Part D

0. Exit

Enter your choice: 0

Exiting...

Result

The menu-driven Java program was successfully implemented using **classes, objects, methods**, `Scanner`, `switch`, `if-else`, `for` **loops**, `while` **loops**, and **arithmetic operations**. The program successfully performs all the operations provided in Parts A, B, C, and D.